Bàrbara Barceló-Llull

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CV summary

Bàrbara Barceló-Llull obtained her PhD in Physical Oceanography at the U. de Las Palmas de Gran Canaria (Spain) in September 2017 with Cum Laude and International PhD mentions. After her PhD, she was a postdoctoral researcher at IMEDEA (Spain) in 2018 and at the U. of Washington (UW, USA) from December 2018 until June 2020. The UW is recognized as one of the world's best universities in Oceanography, ranked 1st between 2018 and 2021 from the ShanghaiRanking's global ranking. Since June 2020, she has continued her postdoctoral research at IMEDEA (María de Maeztu Excellence Unit 2023-2027), first as task leader of the H2020 EuroSea project and later leading her own project.

Her research focuses on in situ and satellite ocean observations to understand mesoscale and submesoscale dynamics and their impact on marine ecosystems. These fine-scale processes regulate heat, carbon, and marine organism distributions, and understanding their 3D structure is a major challenge in physical oceanography. She has advanced knowledge of fine-scale ocean dynamics, with applications to satellite oceanography, climate, and conservation. At UW, she developed an innovative Lagrangian method to improve the resolution of sea surface salinity observations from NASA's SMAP satellite, enabling new insights into salinity-driven density fronts. At IMEDEA, she was a key contributor to the SWOT NASA/CNES satellite validation, leading a task of the H2020 EuroSea project to optimize ocean sampling strategies, coordinating international collaborations, and publishing codes and recommendations adopted by the SWOT Science Team. She has also advanced knowledge of fine-scale 3D hydrodynamics, exploiting historical glider observations and in situ multi-platform observations from the PRE-SWOT and PUMP experiments. Since 2020, she has led the only Spanish group participating in the SWOT Science Team (co-PI VERSO project). Her codes and recommendations have contributed to plan regional SWOT validation, including the FaSt-SWOT experiments.

She has secured competitive funding as Principal Investigator for three projects (∼240.000€) : 'Connectivity of Marine Protected Areas in the Balearic Islands' (2023–2026), funded by the Vicenç Mut program (120.000€); and two successive editions of 'MonSubHeat: Monitoring the Subsurface Signature of Marine Heatwaves in the Balearic Sea', awarded in 2024 and 2025 through SOCIB's Glider Fleet Open Access calls, granting a total of 98 days of glider time valued at ∼120.000€ for experiments in summer 2025 and summer 2026, respectively. Through this funding, she has initiated a research program focusing on the influence of mesoscale variability on large-scale ocean circulation and climate, the drivers and impacts of extreme oceanic events such as marine heatwaves, and the role of ocean dynamics in biodiversity conservation. Additionally, she will leverage new SWOT satellite observations in Antarctica as part of the Coupling II project, providing unprecedented insights into fine-scale ocean dynamics in this region. This research program contributes to the UN Ocean Decade (2021–2030). In total, she leads 4 projects and has participated in 16 additional projects.

She has published 9 first-author peer-reviewed papers in high-impact Q1 journals, and 11 additional papers and 2 book chapters as co-author (second author in 3 papers and 1 book chapter). She has 2 manuscripts under review. She led 2 deliverables and participated in an additional deliverable for the EU during EuroSea; she contributed to 7 additional reports. She actively promotes open science by publishing datasets and codes in public repositories. She has disseminated her work through 27 communications at national and international conferences, being co-author of 42 more communications. She organized a session in the EGU General Assembly in 2019 and in the Ocean Sciences Meeting in 2025. She was a member of the scientific committee of the VIII Encuentro Oceanografía Física conference in 2024 and of the 57th International Liège Colloquium in 2026. She co-chaired 4 sessions in 2 international conferences. She participated in 4 oceanographic experiments (67 days in total). She supervised 5 master's theses, 3 with JAE Intro fellowships. Among them, 1 is now a data scientist in the private sector, and 3 are pursuing their PhD. One of these PhD students is now under her supervision, funded by an FPU fellowship. She participated in 2 master's thesis evaluation committees at the UIB and assessed a master thesis for the U. Auckland (New Zealand). She has reviewed manuscripts for several journals, such as Nature Communications. She was guest editor of two special issues in Remote Sensing.

During 2020-2021 she was a member of the Commission for communication and scientific dissemination at IMEDEA. She actively participated in organizing and developing outreach activities, including designing the short documentary Ilusiones científicas (2021). She has led workshops and seminars and created a video series on YouTube sharing insights into her life as a scientist with the general public (55.000+ views).

Education

2014 - 2017 **Ph.D., Physical Oceanography**, *Instituto de oceanografía y cambio global, IOCAG*, Universidad de Las Palmas de Gran Canaria, Spain.

Thesis: *'Shedding light into mesoscale dynamics and vertical motion through synthetic and in situ observations'*. Cum Laude and International PhD mentions. Date of defense: 25-09-2017. Supervisors: Pablo Sangrà, Ananda Pascual, Enric Pallàs-Sanz and Ángeles Marrero-Díaz. Pre-doctoral grant from the Spanish National Research Program associated to the PUMP project (BES-2013-065459).

2012 - 2014 **M.Sc., Physics**, Universitat de les Illes Balears, Spain, 120 ECTS.

Thesis: *'Quasi-geostrophic vertical motion from satellite and in-situ observations: impact on South East Pacific nitrate distribution through a Lagrangian simulation'*. Supervisors: Ananda Pascual and Evan Mason

2011 - 2012 **M.Sc., Renewable energies**, Universidad Camilo José Cela, Spain, 71 ECTS.

2006 - 2011 **B.Sc., Physics**, Universidad de Salamanca, Spain.

Professional experience

2023 - present **Postdoctoral grant (Connectivitat de la xarxa d'Àrees Marines Protegides de les Illes Balears project, funded by the Vicenç Mut program from the Balearic Government, PD/008/2022, 120.000€)**, *Institut Mediterrani d'Estudis Avançats, IMEDEA (Unidad de Excelencia María de Maeztu 2023-2027)*, Mallorca, Spain.
Supervisor: Josep Alós

2020 - 2023 **Postdoctoral position (H2020 EuroSea project, funded by the European Commission, BG-07-2019-2020)**, *Institut Mediterrani d'Estudis Avançats, IMEDEA (Unidad de Excelencia María de Maeztu 2023-2027)*, Mallorca, Spain.
Supervisor: Ananda Pascual

2018 - 2020 **Postdoctoral position ((Sub)mesoscale Salinity Variability at Fronts project, funded by the NASA, NNX17AK04G)**, *Applied Physics Laboratory, University of Washington*, Seattle, USA.
Supervisor: Kyla Drushka

2017 - 2018 **Postdoctoral position (PRE-SWOT project, CTM2016-78607-P)**, *Institut Mediterrani d'Estudis Avançats, IMEDEA*, Mallorca, Spain.
Supervisor: Ananda Pascual

2014 - 2017 **Ph.D. student (PUMP project, CTM2012-33355)**, *Universidad de Las Palmas de Gran Canaria*, Las Palmas, Spain.
Supervisors: Pablo Sangrà, Ananda Pascual, Enric Pallàs-Sanz and Ángeles Marrero-Díaz.

2012 - 2013 **Research assistant (E-MOTION project, CTM2012-31014)**, *Institut Mediterrani d'Estudis Avançats, IMEDEA*, Mallorca, Spain.
Supervisor: Ananda Pascual

Research project participation

1. *Name of the project:* RECONECTA: Reservas Conectadas: Hacia una red resiliente de gestión pesquera en Mallorca
PI: Andrés Alonso Ospina Álvarez
Funding entity: Programa Pleamar 2025, Fundación Biodiversidad, Fondo Europeo Marítimo, de Pesca y Acuicultura (FEMPA)
Start-End date: 27/01/2026 - 26/01/2028
Total amount: 222.621€

2. *Name of the project:* MASAGEH: Mesoscale and (sub)mesoscale activity in the La Gomera and El Hierro islands corridor: Impacts on regional dynamics and biogeochemistry (PID2024-161413NA-I00)
   PI: Yeray Santana Falcón
   Funding entity: Spanish National Research Program and European Regional Development Fund (MINECO-FEDER)
   Start-End date: 01/09/2025 - 31/08/2028
   Total amount: 106.250€

3. *Name of the project:* MonSubHeat: Monitoring the subsurface signature of Marine Heatwaves in the Balearic Sea (0053)
   PI: **Bàrbara Barceló-Llull**
   Funding entity: Balearic Islands Coastal Observing and Forecasting System (SOCIB)
   Call: Glider Fleet Competitive Open Access (SOCIB 2025 N.1 AAC)
   Link to the call: `https://www.socib.es/en/what-we-do/access/access-to-the-glider-fleet`
   Start-End date: 12/01/2026 - 15/12/2026
   Total value of access granted: 64.579,44€ (53 days of free access to the glider, valued at 1.218,48€/day)

4. *Name of the project:* Relationship between the Atlantic meridional overturning circulation and coastal sea level: New insights from next-generation satellite observations
   PI: Christopher G. Piecuch (WHOI, USA), and Julius Oelsmann (Tulane University, USA)
   Funding entity: International Space Science Institute (ISSI) in Bern, Switzerland
   Call: ISSI/ISSI-BJ 2025 Joint Call for Proposals for International Teams in the Space & Earth Sciences
   Link to the call: `https://www.issibern.ch/call-for-proposals-2025-for-international-teams-in-space-and-earth-sciences/`
   Start-End date: 13/06/2025 - 13/06/2027
   Total amount: The ISSI supports two 5-day International Team meetings in Bern for all 12 core Team members as well as two additional early-career researchers, covering accommodation, local expenses, meeting infrastructure, childcare support, and travel for Team leaders.

5. *Name of the project:* MonSubHeat: Monitoring the subsurface signature of Marine Heatwaves in the Balearic Sea (0044)
   PI: **Bàrbara Barceló-Llull**
   Funding entity: Balearic Islands Coastal Observing and Forecasting System (SOCIB)
   Call: Glider Fleet Competitive Open Access (SOCIB 2024 N.2 AAC)
   Link to the call: `https://www.socib.es/en/what-we-do/access/access-to-the-glider-fleet`
   Start-End date: 15/01/2025 - 15/12/2025
   Total value of access granted: 54.831,60€ (45 days of free access to the glider, valued at 1.218,48€/day)
6. *Name of the project:* Coupling II: Ocean Dynamics and Connectivity between the Bransfield Current System and the Western Boundary Current System of the Weddell Sea (PID2023-148583NB-C21)
   PIs: Miguel Borja Aguiar González; Francisco José Machín Jiménez
   Project website: `https://polarcoupling.com/`
   Funding entity: Spanish National Research Program and European Regional Development Fund (MINECO-FEDER)
   Start-End date: 01/09/2024 - 31/08/2028
   Total amount: 257.500€
7. *Name of the project:* Ocean observations and indicators for climate and assessments (ObsSea4Clim) (No. 101136548)
   PIs: S. Olsen, V. Combes
   Funding entity: European Research Executive Agency
   Call: HORIZON-CL6-2023-CLIMATE-01
   Start-End date: 01/02/2024- 31/01/2028
   Total amount: 5.996.902€
8. *Name of the project:* RATJADA: Contribución al efecto de las Áreas Marinas Protegidas en el mantenimiento de la biodiversidad de condrictios batoideos de las Islas Baleares utilizando marcado electrónico (GOIBE299455/2024 BIOPRO2024-009)
   PI: Josep Alós Crespí
   Funding entity: Balearic Government
   Start-End date: 01/01/2024 - 29/09/2025
   Total amount: 116.900€

9. *Name of the project:* Connectivitat de la xarxa d'Àrees Marines Protegides de les Illes Balears (PD/008/2022)
   PI: **Bàrbara Barceló-Llull**
   Funding entity: Vicenç Mut program from the Balearic Government
   Start-End date: 01/12/2023 - 30/11/2026
   Total amount: 120.000€
10. *Name of the project:* VERSO: FINE SCALE [VER]TICAL EXCHANGE IN THE UPPER OCEAN – [S]W[O]T (continuation)
   PIs: Ananda Pascual; Baptiste Mourre; Antonio Sánchez Román; **Bàrbara Barceló-Llull**; Laura Gómez-Navarro
   Funding entity: SWOT Science Team - CNES/NASA/EUMETSAT
   Start-End date: 01/05/2024 - 01/05/2027
11. *Name of the project:* METARAOR: Population and managerial consequences of fish behavioral variation in marine fish species (PID2022-139349OB-I00)
   PI: Josep Alós Crespí
   Funding entity: Spanish National Research Program and European Regional Development Fund (MINECO-FEDER)
   Start-End date: 01/09/2023 - 31/08/2026
   Total amount: 303.750€
12. *Name of the project:* 4DMED-Sea (CNR-ESA-2022-4DMED-SEA)
   PIs: B. Buongiorno Nardelli; I. Hernández-Carrasco
   Funding entity: European Space Agency
   Call: ESA AO/1-11311/22/I-DT EXPRO+
   Website: `http://ricerca.ismar.cnr.it/4DMED/index.html`
   Start-End date: 22/06/2023 - 21/06/2025
   Total amount: 699.869,40 €
13. *Name of the project:* FaSt-SWOT: Fine-Scale ocean currents from integrated multi-platform experiments and numerical simulations: contribution to the new SWOT satellite mission (PID2021-122417NB-I00)
   PIs: Ananda Pascual; Baptiste Mourre
   Funding entity: Spanish National Research Program and European Regional Development Fund (MINECO-FEDER)
   Start-End date: 01/09/2022 - 31/08/2026
   Total amount: 198.440€
14. *Name of the project:* VERSO: FINE SCALE [VER]TICAL EXCHANGE IN THE UPPER OCEAN – [S]W[O]T
   PIs: Ananda Pascual; Evan Mason; Antonio Sánchez Román; **Bàrbara Barceló-Llull**; Ronan Fablet
   Funding entity: SWOT Science Team - CNES/NASA/EUMETSAT
   Start-End date: 01/05/2020 - 01/05/2024

15. *Name of the project:* EUROSEA: IMPROVING AND INTEGRATING EUROPEAN OCEAN OBSERVING AND FORECASTING SYSTEMS FOR SUSTAINABLE USE OF THE OCEANS (BG-07-2019-2020)
    PIs: Toste Tanhua; Ananda Pascual
    Funding entity: European Commission (H2020)
    Start-End date: 01/11/2019 - 30/11/2023
    Total amount: 12.000.000€
16. *Name of the project:* (Sub)mesoscale Salinity Variability at Fronts (NNX17AK04G)
    PI: Kyla Drushka
    Funding entity: National Aeronautics and Space Administration (NASA)
    Start-End date: 20/04/2017 - 19/04/2020
    Total amount: $99.595
17. *Name of the project:* PRE-SWOT: INTERCAMBIOS VERTICALES DE MESOESCALA Y SUBMESOESCALA A PARTIR DE EXPERIMENTOS MULTI-PLATAFORMA Y SIMULACIONES NUMERICAS: ACTIVIDADES PREVIAS AL LANZAMIENTO DEL SATELITE SWOT (CTM2016-78607-P)
    PIs: Ananda Pascual; John Allen
    Funding entity: Spanish National Research Program and European Regional Development Fund (MINECO-FEDER)
    Start-End date: 01/01/2017 - 31/12/2019
    Total amount: 146.000€
18. *Name of the project:* MULTI-SUB: MESOSCALE AND SUB-MESOSCALE VERTICAL EXCHANGES FROM MULTI-PLATFORM EXPERIMENTS AND SUPPORTING MODELING SIMULATIONS: ANTICIPATING SWOT LAUNCH
    PIs: Ananda Pascual
    Funding entity: SWOT Science Team - CNES/NASA/EUMETSAT
    Start-End date: 01/12/2015 - 01/12/2019
19. *Name of the project:* FLUXES: FLUJOS DE CARBONO EN UN SISTEMA DE AFLORAMIENTO COSTERO (CABO BLANCO, NW DE AFRICA); MODULACION A SUBMESOSCALA DE LA PRODUCCION, EXPORTACION Y CONSUMO DE CARBONO (CTM2015- 69392-C3-3-R)
    PIs: Ángel Rodríguez Santana; Antonio Martínez Marrero
    Funding entity: Spanish National Research Program and European Regional Development Fund (MINECO-FEDER)
    Start-End date: 01/01/2016 a 31/12/2019
    Total amount: 139.150€

20. *Name of the project:* E-MOTION: Ocean mesoscale Eddies and vertical MOTION: new perspectives from a combination of satellite, model and in situ data (CTM2012-31014)
PIs: Ananda Pascual
Funding entity: Spanish National Research Program
Start-End date: 01/01/2013 - 31/12/2015
Total amount: 38.000€
21. *Name of the project:* PUMP: ESTUDIO DE LA BOMBA VERTICAL OCEANICA EN REMOLINOS DE MESOSCALA (CTM2012-33355)
PIs: Pablo Sangrà Inciarte
Funding entity: Spanish National Research Program
Start-End date: 01/01/2013 - 31/12/2015
Total amount: 222.300€

Publications

1. Cortés-Morales, D., **Barceló-Llull, B.**, Gómez-Navarro, L. and Pascual, A. (2026, under review). Eddy Kinetic Energy Contribution to Total Kinetic Energy in the Global Ocean from 23 Years of Multi-Satellite Altimetry. *Geophysical Research Letters*.
2. Fernández-Álvarez, B., **Barceló-Llull, B.** and Pascual, A. (2025, under review). Mediterranean Sea Warming and Marine heatwaves in 2024. *State of the Planet Discussions*. DOI: `https://doi.org/10.5194/sp-2025-13`
3. Hargous, P., Combes, V., **Barceló-Llull, B.** and Pascual, A. (2026, accepted). Eddy Kinetic Energy Variability From 30 Years of Altimetry in the Mediterranean Sea. *EGUsphere [preprint]*. DOI: `https://doi.org/10.5194/egusphere-2025-4651`
4. Fernández-Álvarez, B., **Barceló-Llull, B.** and Pascual, A. (2025). Tracking marine heatwaves in the Balearic Sea: temperature trends and the role of detection methods. *Ocean Sciences*, 21, 1987–1999. DOI: `https://doi.org/10.5194/os-21-1987-2025`
5. Verger-Miralles, E., Mourre, B., Gómez-Navarro, L., **Barceló-Llull, B.**, Casas, B., Cutolo, E., Díaz-Barroso, L., d'Ovidio, F., Tarry, D. R., Zarokanellos, N. D. and Pascual, A. (2025). SWOT enhances small-scale eddy detection in the Mediterranean Sea. *Geophysical Research Letters*, 52, e2025GL116480. DOI: `https://doi.org/10.1029/2025GL116480`
6. **Barceló-Llull, B.**, Rosselló, P., Combes, V., Sánchez-Román, A., Pujol, M. I. and Pascual, A. (2025). Kuroshio Extension and Gulf Stream dominate the Eddy Kinetic Energy intensification observed in the Global Ocean. *Sci. Rep.*, 15, 21754. DOI: `https://doi.org/10.1038/s41598-025-06149-9`

7. Edwards, C. A., De Mey-Frémaux, P., **Barceló-Llull, B.**, Charria, G., Choi, B.-J., Halliwell, G. R., Hole, L. R., Kerry, C., Kourafalou, V. H., Kurapov, A. L., Moore, A. M., Mourre, B., Oddo, P., Pascual, A., Roughan, M., Skandrani, C., Storto, A., Vervatis, V. and Wilkin, J. L. (2024). Assessing impacts of observations on ocean circulation models with examples from coastal, shelf, and marginal seas. *Front. Mar. Sci.*, Volume 11. DOI: `10.3389/fmars.2024.1458036`

8. Tanhua, T., Le Traon, P.-Y., Köstner, N., Eparkhina, D., Navarro, G., Bonnet Dunbar, M., Speich, S., Pascual, A., Von Schuckmann, K., Liguori, G., Karstensen, J., Hassoun, A. E. R., Van Doorn, E., **Barceló-Llull, B.**, Pérez-Gómez, B., Cusack, C., Heslop, E., Lara-Lopez, A., Petihakis, G., Telszewski, M., Piotr Palacz, A., Wilmer-Becker, K., Pearlman, J. S., Muñiz Piniella, Á., Heymans, J. J. and Lips, I. (2024). Towards a sustained and fit-for-purpose European Ocean Observing and Forecasting System. *Front. Mar. Sci.*, 11:1394549. DOI: `https://doi.org/10.3389/fmars.2024.1394549`

9. **Barceló-Llull, B.** and Pascual, A. (2023). Recommendations for the design of in situ sampling strategies to reconstruct fine-scale ocean currents in the context of SWOT satellite mission. *Front. Mar. Sci.*, 10:1082978. DOI: `https://doi.org/10.3389/fmars.2023.1082978`

10. Mason, E., **Barceló-Llull, B.**, Sánchez-Román, A., Rodríguez-Tarry, D., Cutolo, E., Delepoulle, A., Ruiz, S. and Pascual, A. (2022). Chapter 8: Fronts, eddies and mesoscale circulation in the Mediterranean Sea. Book title: Oceanography of the Mediterranean Sea. An Introductory Guide. Editors: Schroeder, K. and Chiggiato, J. Publisher: *Elsevier*. ISBN: 978-0-12-823692-5

11. Tzortzis, R., Doglioli, A. M., Barrillon, S., Petrenko, A. A., d'Ovidio, F., Izard, L., Thyssen, M., Pascual, A., **Barceló-Llull, B.**, Cyr, F., Tedetti, M., Bhairy, N., Garreau, P., Dumas, F., and Gregori, G. (2021). Impact of moderately energetic fine-scale dynamics on the phytoplankton community structure in the western Mediterranean Sea. *Biogeosciences*, 18, 6455-6477. DOI: `https://doi.org/10.5194/bg-18-6455-2021`

12. **Barceló-Llull, B.**, Pascual, A., Sánchez-Román, A., Cutolo, E., d'Ovidio, F., Fifani, G., Ser-Giacomi, E., Ruiz, S., Mason, E., Cyr, F., Doglioli, A., Mourre, B., Allen, J. T., Alou-Font, E., Casas, B., Díaz-Barroso, L., Dumas, F., Gómez-Navarro, L. and Muñoz, C. (2021). Fine-Scale Ocean Currents Derived From in situ Observations in Anticipation of the Upcoming SWOT Altimetric Mission. *Front. Mar. Sci.*, 8:679844. DOI: `https://doi.org/10.3389/fmars.2021.679844`

13. International Altimetry Team (2021). Altimetry for the future: Building on 25 years of progress. *Advances in Space Research*, 68(2), 319-363. DOI: `https://doi.org/10.1016/j.asr.2021.01.022`

14. **Barceló-Llull, B.**, Drushka, K. and Gaube, P. (2021). Lagrangian reconstruction to extract small-scale salinity variability from SMAP observations. *Journal of Geophysical Research: Oceans*, 126, e2020JC016477. DOI: `https://doi.org/10.1029/2020JC016477`

15. Martínez-Marrero, A., **Barceló-Llull, B.**, Pallàs-Sanz, E., Aguiar-González, B., Estrada-Allis, S. N., Gordo, C., Grisolía, D., Rodríguez-Santana, A. and Arístegui, J. (2019). Near-inertial wave trapping near the base of an anticyclonic mesoscale eddy under normal atmospheric conditions. *Journal of Geophysical Research: Oceans*, 124(11), 8455-8467. DOI: `https://doi.org/10.1029/2019JC015168`

16. **Barceló-Llull, B.**, Pascual, A., Ruiz, S., Escudier, R., Torner, M. and Tintoré, J. (2019). Temporal and spatial hydrodynamic variability in the Mallorca channel (western Mediterranean Sea) from eight years of underwater glider data. *Journal of Geophysical Research: Oceans*, 124(4), 2769-2786. DOI: `https://doi.org/10.1029/2018JC014636`

17. Estrada-Allis, S. N., **Barceló-Llull, B.**, Pallàs-Sanz, E., Rodríguez-Santana, A., Souza, J. M. A. C., Mason, E., McWilliams, J. C. and Sangrà, P. (2019). Vertical Velocity Dynamics and Mixing in an Anticyclone near the Canary Islands. *J. Phys. Oceanogr.*, 49(2), 431-451, DOI: `https://doi.org/10.1175/JPO-D-17-0156.1`

18. **Barceló-Llull, B.**, Pascual, A., Mason, E. and Mulet, S. (2018). Comparing a Multivariate Global Ocean State Estimate With High-Resolution *in Situ* Data: An Anticyclonic Intrathermocline Eddy Near the Canary Islands. *Front. Mar. Sci.*, 5(66), DOI: `https://doi.org/10.3389/fmars.2018.00066`

19. **Barceló-Llull, B.**, Pallàs-Sanz, E., Sangrà, P., Martínez-Marrero, A., Estrada-Allis, S. N. and Arístegui, J. (2017). Ageostrophic secondary circulation in a subtropical intrathermocline eddy. *J. Phys. Oceanogr.*, 47(5), 1107-1123, DOI: `http://dx.doi.org/10.1175/JPO-D-16-0235.1`

20. **Barceló-Llull, B.**, Sangrà, P., Pallàs-Sanz, E., Barton, E. D., Estrada-Allis, S. N., Martínez-Marrero, A., Aguiar-González, B., Grisolía, D., Gordo, C., Rodríguez-Santana, A., Marrero-Díaz, A. and Arístegui, J. (2017). Anatomy of a subtropical intrathermocline eddy. *Deep-Sea Res.*, 124, 126-139, DOI: `http://dx.doi.org/10.1016/j.dsr.2017.03.012`. *Selected for the front cover of Volume 124 by the editor Igor Belkin. `https://doi.org/10.1016/S0967-0637(17)30172-3`

21. **Barceló-Llull, B.**, Mason, E., Capet, A. and Pascual, A. (2016). Impact of vertical and horizontal advection on nutrient distribution in the southeast Pacific. *Ocean Sci.*, 12, 1003-1011, DOI: `http://dx.doi.org/10.5194/os-12-1003-2016`

22. Rio, M.-H., Pascual, A., Poulain, P.-M., Menna, M., **Barceló, B.** and Tintoré, J. (2014). Computation of a new Mean Dynamic Topography for the Mediterranean Sea from model outputs, altimeter measurements and oceanographic in-situ data. *Ocean Sci.*, 10, 731-744, DOI: `http://dx.doi.org/10.5194/os-10-731-2014`

23. Tintoré, J. *et al.* (2013). The Impact of New Multi-platform Observing Systems in Science, Technology Development and Response to Societal Needs; from Small to Large Scales. *Computer Aided Systems Theory - EUROCAST 2013*, Springer Berlin Heidelberg, 8112, 341-348, DOI: `http://dx.doi.org/10.1007/978-3-642-53862-9_44`
24. Tintoré, J. *et al.* (2013). SOCIB: The Balearic Islands Coastal Ocean Observing and Forecasting System Responding to Science, Technolog y and Society Needs. *Mar. Technol. Soc. J.*, 47(1), 101-117, DOI: `http://dx.doi.org/10.4031/MTSJ.47.1.10`

Reports

1. Barrientos, N., Vaquer-Sunyer, R., Juza, M., Vargas-Yáñez, M., Gomis, D., Pascual, A., **Barceló-Llull, B.**, Balbín, R., Jordà, G. and Marcos, M. (2024). Temperatura de la mar Balear. In: Vaquer-Sunyer, R.; Barrientos, N.; Gouraguine, A. (eds.). Informe Mar Balear 2024, DOI: `http://doi.org/10.62135/IAJB7896`
2. Barrientos, N., Vaquer-Sunyer, R., Juza, M., Vargas-Yáñez, M., Gomis, D., Pascual, A., **Barceló-Llull, B.**, Balbín, R., Jordà, G. and Marcos, M. (2024). Temperatura del mar Balear. In: Vaquer-Sunyer, R.; Barrientos, N.; Gouraguine, A. (eds.). Informe Mar Balear 2024, DOI: `https://doi.org/10.62135/LDBJ9344`
3. Mourre, B., Pascual, A., Gómez-Navarro, L., Balaguer, P., **Barceló-Llull, B.**, *et al.* (2024). FaSt-SWOT 2023 Cruise Report. FaSt-SWOT project, DOI: `https://doi.org/10.20350/digitalCSIC/16077`
4. Pascual, A., Mourre, B., Balaguer, P., **Barceló-Llull, B.**, *et al.* (2023). FaSt-SWOT Cruise Plan: Fine-Scale ocean currents from integrated multi-platform experiments and numerical simulations: contribution to the new SWOT satellite mission (FaSt-SWOT, PID2021-122417NB-I00). FaSt-SWOT project, 56pp., DOI: `https://doi.org/10.20350/digitalCSIC/15276`
5. Speich, S., Pascual, A., Dayan, H., **Barceló-Llull, B.**, Balmaseda, M., Cusack, C., Dabrowski, T., Graves, I., Horsburgh, K., Lips, U., McAdam, R., Masina, S., Navarro, G., She, J. & von Schuckmann, K. (2022). Development of targeted indicators and their uncertainties for demonstrators and Forecasts. EuroSea project, Deliverable 2.4, DOI: `https://doi.org/10.3289/eurosea_d2.4`
6. Mulet, S., Boone, C., **Barceló-Llull, B.**, Zarokanellos, N., Pascual, A., Manso-Narvarte, I., Caballero, A., Rubio, A., Bretagnon, M. & Mangin, A. (2022). Synthesis of satellite validation results. EuroSea project, Deliverable 4.5, DOI: `https://doi.org/10.3289/eurosea_d4.5`

7. **Barceló-Llull, B.**, Pascual, A., Albert, A., Beauchamp, M., Fablet, R., Guinehut, S., Hernández-Lasheras, J., Herrero González, M., Jousset, S., Leroux, S., Mourre, B. & Mulet, S. (2022). Analysis of the OSSEs with multi-platform in situ data and impact on fine-scale structures. EuroSea project, Deliverable 2.3, DOI: `https://doi.org/10.3289/eurosea_d2.3_v2`
8. Barrientos, N., Vaquer-Sunyer, R., Gomis, D., Marcos, M., Jordà, G., **Barceló-Llull, B.**, Pascual, A. and Ruiz-Parrado, I. (2021). Salinidad. In: Vaquer-Sunyer, R.; Barrientos, N. (ed.). Informe Mar Balear 2021, DOI: `https://doi.org/10.62135/ECDV8493`
9. **Barceló-Llull, B.**, Pascual, A., Cutolo, E., Fablet, R., Gasparin, F., Guinehut, S., Hernández-Lasheras, J., Leroux, S., Mignot, A., Mourre, B., Mulet, S., Rémy, E., Speich, S. & Verbrugge, N. (2020). Design of the Observing System Simulation Experiments with multi-platform in situ data and impact on fine-scale structures. EuroSea project, Deliverable 2.1, DOI: `https://doi.org/10.3289/eurosea_d2.1`
10. **Barceló-Llull, B.**, Pascual, A., *et al.* (2018). PRE-SWOT Cruise Report: Mesoscale and sub-mesoscale vertical exchanges from multi-platform experiments and supporting modeling simulations: anticipating SWOT launch. PRE-SWOT project, 145pp., DOI: `http://dx.doi.org/10.20350/digitalCSIC/8584`

Datasets and codes in public repositories

1. **Barceló-Llull, B.** & Rosselló, P. (2025). Code accompanying the paper "Kuroshio Extension and Gulf Stream dominate the Eddy Kinetic Energy intensification observed in the global ocean" published in Scientific Reports (2025) (v1.0.0). *Zenodo*, DOI: `https://doi.org/10.5281/zenodo.15655616`
2. Mourre, B., Gómez-Navarro, L., Verger-Miralles, E., Casas, B., **Barceló-Llull, B.**, *et al.* (2024). FaSt-SWOT multiplatform experiment raw dataset [Dataset]. *DIGITAL.CSIC*, DOI: `https://doi.org/10.20350/digitalCSIC/16511`
3. **Barceló-Llull, B.** (2023). Codes to reconstruct Sea Surface Salinity observations made by the Soil Moisture Active Passive (SMAP) satellite in the northwest Atlantic Ocean. *GitHub*, URL: `https://github.com/bbarcelollull/SSS_reconstruction_Gulf_Stream`
4. **Barceló-Llull, B.** and Verger-Miralles, E. (2023). Codes to design ocean sampling strategies. *GitHub*, URL: `https://github.com/bbarcelollull/design_ocean_sampling_strategies`
5. **Barceló-Llull, B.** (2023). Codes generated to evaluate in situ sampling strategies to reconstruct fine-scale ocean currents in the context of SWOT satellite mission (H2020 EuroSea project) (v1.0.0). *Zenodo*, DOI: `https://doi.org/10.5281/zenodo.7543697`

6. **Barceló-Llull, B.**, Pascual, A., *et al.* (2022). Dataset generated to evaluate in situ sampling strategies to reconstruct fine-scale ocean currents in the context of SWOT satellite mission (H2020 EuroSea project). *Zenodo*, DOI: `https://doi.org/10.5281/zenodo.6798018`
7. Pascual, A., **Barceló-Llull, B.**, *et al.* (2019). Dataset from PRE-SWOT multi-platform experiment. *DIGITAL.CSIC*, DOI: `http://dx.doi.org/10.20350/digitalCSIC/8640`

Conferences

1. Hargous, P., Combes, V., **Barceló-Llull, B.** and Pascual, A. (2025). Long-term variability of eddy kinetic energy in the Mediterranean Sea: A multi-product altimetry perspective. *MonGOOS workshop*, Lisbon, Portugal. Oral. 3–4 December 2025.
2. Pascual, A., Mourre, B., Verger-Miralles, E., Gómez-Navarro, L., **Barceló-Llull, B.**, Tarry, D. R., Cutolo, E., López, G., Reyes, E. and Zarokanellos, N. (2025). Mapping the Known and Invisible in the age of SWOT: Fine-Scale Ocean Currents from the VERSO/FaSt-SWOT collaboration. *SWOT Science Team Meeting*, Arcachon, France. Oral. 14–17 October 2025.
3. Gómez-Navarro, L., Tarry, D. R., Verger-Miralles, E., Cortes-Morales, D., **Barceló-Llull, B.**, Vega-Gimenez, D., Zarokanellos, N., Diaz-Barroso, L., Reyes, E., Lizaran, I., Mourre, B. and Pascual, A. (2025). High-Resolution Observations from the FaSt-SWOT Campaigns: Validation and Fine-Scale Signal Analysis. *SWOT Science Team Meeting*, Arcachon, France. Oral. 14–17 October 2025.
4. Mourre, B., Peng, J.-P., Verger-Miralles, E., Gómez-Navarro, L., García-Jove, M., Cutolo, E., **Barceló-Llull, B.**, Tarry, D. R., and Zarokanellos, N. (2025). Assessment of SWOT satellite observations and integration with high-resolution regional simulations in the Western Mediterranean Sea. *SWOT Science Team Meeting*, Arcachon, France. Poster. 14–17 October 2025.
5. Combes, V., Martí-Solana, C., and **Barceló-Llull, B.** (2025). Detecting Salinity Fronts from Satellite Observations using a Lagrangian reconstruction method. *Parcels Workshop-Conference*, Utrecht, Netherlands (hybrid). Oral. 1–3 October 2025.
6. Verger-Miralles, E., Mourre, B., Gómez-Navarro, L., **Barceló-Llull, B.**, Cutolo, E., Tarry, D. R., Zarokanellos, N. D. and Pascual, A. (2025). Enhanced detection and reconstruction of a small-scale intrathermocline eddy using SWOT and high-resolution in-situ data in the Western Mediterranean. *ESA Living Planet Symposium*, Vienna, Austria. Oral. 23-27 June 2025.

7. Gómez-Navarro, L., Verger-Miralles, E., Tarry, D. R., **Barceló-Llull, B.**, Zarokanellos, N., Diaz-Barroso, L., Lopez, G., Lizaran, I., Reyes, E., Mourre, B. and Pascual, A. (2025). SWOT validation and analyses during the fast-sampling phase in the Western Mediterranean Sea with high-resolution observations: assessing SWOT satellite performance against tide-gauge observations. *ESA Living Planet Symposium*, Vienna, Austria. Oral. 23-27 June 2025.
8. Hargous, P., Combes, V., **Barceló-Llull, B.** and Pascual, A. (2025). Eddy Kinetic Energy Intensification in the Mediterranean Sea From Three Decades of Satellite Altimetry Observations. *ESA Living Planet Symposium*, Vienna, Austria. Poster. 23-27 June 2025.
9. Mourre, B., Verger-Miralles, E., Peng, J.-P., Gómez-Navarro, L., Garcia-Jove, M., Cutolo, E., **Barceló-Llull, B.**, Tarry, D. R., Zarokanellos, N. and Pascual, A. (2025). SWOT satellite sea level observations: validation and integration with high-resolution regional simulations. *COSS-TT Meeting*, Ifremer, Plouzané, France. Oral. 17-20 June 2025.
10. Hargous, P., Combes, V., **Barceló-Llull, B.**, and Pascual, A. (2025). Satellite Altimetry Reveals Intensification of Eddy Kinetic Energy in the Mediterranean Sea. *EGU General Assembly*, Vienna, Austria. Oral. 27 Apr–2 May 2025.
11. Combes, V., Martí-Solana, C., and **Barceló-Llull, B.** (2025). Detecting Salinity Fronts From Satellite Observations in the Brazil-Malvinas Confluence. *EGU General Assembly*, Vienna, Austria. Oral. 27 Apr–2 May 2025.
12. Fernández-Álvarez, B., **Barceló-Llull, B.**, and Pascual, A. (2025). Tracking Marine Heatwaves in the Balearic Sea: Temperature Trends and the Role of Detection Methods. *EGU General Assembly*, Vienna, Austria. Poster. 27 Apr–2 May 2025.
13. Mourre B., Verger-Miralles, E., Gómez-Navarro, L., Garcia-Jove, M., **Barceló-Llull, B.**, Casas, B., Combes, V., Cutolo, E., Díaz-Barroso, L., Lizarán, I., Reyes, E., Tarry, D. R., Zarokanellos, N., and Pascual, A. (2024). SWOT satellite sea level observations: a game-changer for high- resolution ocean prediction? Insights from a Western Mediterranean Sea experiment. *Ocean Predict Symposium*, Paris, France. Poster. 18-21 November 2024.
14. **Barceló-Llull, B.**, Catalán, I. A., Ospina-Álvarez, A., Barcelo-Serra, M., Cortès Pujol, M. À., Aspillaga, E. and Alós, J. (2024). Addressing Connectivity Networks in Marine Protected Areas using Hydrodynamic Modeling and Acoustic Telemetry. *European Tracking Network Symposium*, Mallorca, Spain. Poster. 16-18 October 2024.

15. Alós, J., Abad-Platas, J., Aspillaga, E., **Barceló-Llull, B.** and Nicolau Llobera, M. C. (2024). Towards long-term (>10 years) acoustic monitoring of endangered batoid chondrichthyans in the Balearic Islands. *European Tracking Network Symposium*, Mallorca, Spain. Poster. 16-18 October 2024.
16. Mourre B., Verger-Miralles, E., Gómez-Navarro, L., **Barceló-Llull, B.**, Casas, B., Combes, V., Cutolo, E., Díaz-Barroso, L., Lizarán, I., Reyes, E., Tarry, D. R., Zarokanellos, N. and Pascual, A. (2024). New insights into small-scale ocean dynamics from SWOT satellite measurements, in situ observations and high-resolution modelling in the Balearic Sea. *MonGOOS workshop "Boost the understanding of the Mediterranean for adaptation to climate change and extreme events"*, Málaga, Spain. Oral. 2 October 2024.
17. **Barceló-Llull, B.**, Rosselló, P., Combes, V., Sánchez-Román, A., Pujol, M. I. and Pascual, A. (2024). Global trends in Eddy Kinetic Energy over the altimetric era: are the oceans becoming more energetic? *30 Years of Progress in Radar Altimetry Symposium*, Montpellier, France. Oral. 2-7 September 2024.
18. Gomez-Navarro, L., Verger-Miralles E., **Barceló-Llull, B.**, Casas B., Combes V., Cutolo E., Diaz-Barroso L., Garcia-Jove M., Lizarrán I., Reyes E., Tarry D., Zarokanellos N., Mourre B., Pascual A. (2024). SWOT validation in the western Mediterranean Sea with high-resolution observations. *30 Years of Progress in Radar Altimetry Symposium*, Montpellier, France. Oral. 2-7 September 2024.
19. Verger-Miralles, E., Mourre, B., Gómez-Navarro, L., **Barceló-Llull, B.**, Cutolo, E., Tarry, D., Zarokanellos, N. and Pascual, A. (2024). Analysis of fine-scale dynamics in the Balearic Sea through high-resolution in-situ observations, modelling and SWOT satellite data. *30 Years of Progress in Radar Altimetry Symposium*, Montpellier, France. Poster. 2-7 September 2024.
20. **Barceló-Llull, B.**, Rosselló, P., Combes, V., Sánchez-Román, A., Pujol, M. I. and Pascual, A. (2024). GLOBAL TRENDS IN EDDY KINETIC ENERGY FROM ALTIMETRY: ARE THE OCEANS BECOMING MORE ENERGETIC? *VIII Encuentro de Oceanografía Física (EOF)*, Valencia, Spain. Oral. 10-12 July 2024.
21. Pascual, A., Mourre, B., Gómez-Navarro, L., Verger-Miralles, E., Vega-Giménez, D., Sánchez-Román, A., **Barceló-Llull, B.** (presenter), Casas, B., Combes, V., Cutolo, E., Díaz-Barroso, L., Lizarán, I., Reyes, E., Tarry, D. R. and Zarokanellos, N. (2024). FAST-SWOT: FINE-SCALE OCEAN CURRENTS FROM INTEGRATED MULTI-PLATFORM EXPERIMENTS AND NUMERICAL SIMULATIONS: CONTRIBUTION TO THE NEW SWOT SATELLITE MISSION. *VIII Encuentro de Oceanografía Física (EOF)*, Valencia, Spain. Oral. 10-12 July 2024.
22. Fernández-Álvarez, B., **Barceló-Llull, B.** and Pascual, A. (2024). MARINE HEATWAVES IN THE BALEARIC SEA *VIII Encuentro de Oceanografía Física (EOF)*, Valencia, Spain. Poster. 10-12 July 2024.

23. Verger-Miralles, E., Mourre, B., **Barceló-Llull, B.**, *et al.* (2024). Study of fine-scale dynamics in the Balearic Sea through high-resolution observations and SWOT satellite data. *Marine Technology (MARTECH) Workshop*, Palma, Spain. Oral. 6-7 June 2024.

24. Verger-Miralles, E., Mourre, B., **Barceló-Llull, B.**, Gómez-Navarro, L., Tarry, D. R., Zarokanellos, N. & Pascual, A. (2024). Analysis of fine-scale dynamics in the Balearic Sea through high-resolution observations and SWOT satellite data. *EGU General Assembly 2024*, Vienna, Austria. Oral. 14-19 April 2024.

25. Mourre, B., Verger-Miralles, E., Gómez-Navarro, L., **Barceló-Llull, B.**, Casas, B., Combes, V., Cutolo, E., Diaz-Barroso, L., Garcia-Jove, M., Lizarrán, I., Reyes, E., Tarry, D., Zarokanellos, N., & Pascual, A. (2024). FaSt-SWOT field campaigns combining high-resolution observations and modelling for SWOT validation in the Western Mediterranean Sea. *EGU General Assembly 2024*, Vienna, Austria. Poster. 14-19 April 2024.

26. Martellucci, R., Bellacicco, M., Menna, M., Berta, M., Joel, A., Pacciaroni, M., Mauri, E., Notarstefano, G., Poulain, P. M., Centurioni, L., Bosse, A., Petrenko, A., Arnauld, M., Bouruet-Aubertot, P., **Barceló-Llull, B.**, Tzortzis, R., Rousselet, L., Rolland, R., Gastauer, S., Ohman, M. D., d'Ovidio, F., & Doglioli, A. (2024). Integrating SWOT with autonomous platforms: the case of the BioSWOT-Med biogeochemical front. *EGU General Assembly 2024*, Vienna, Austria. Poster. 14-19 April 2024.

27. Rolland, R., Bouruet-Aubertot, P., Cuypers, Y., Pulido, E., Bosse, A., Petrenko, A., Nunige, S., Rousselet, L., Barrillon, S., Berta, M., Arnaud, M., Menna, M., Pacciaroni, M., Tzortzis, R., **Barceló-Llull, B.**, d'Ovidio, F., Grégori, G., & Doglioli, A. (2024). Internal wave–eddy interactions, turbulence and mixing during the 2023 BioSWOT-MED cruise. *EGU General Assembly 2024*, Vienna, Austria. Poster. 14-19 April 2024.

28. Cutolo, E., Casas, B., **Barceló-Llull, B.**, *et al.* (2024). FaSt-SWOT Experiments: Combining Traditional and Deep-Learning Approaches for Fine-Scale Ocean Sampling and 3D Reconstruction to Validate SWOT in the Western Mediterranean. *Ocean Sciences meeting*, New Orleans, USA. Oral. 23 February 2024.

29. Pascual, A., Mourre, B., Balaguer, P., **Barceló-Llull, B.**, *et al.* (2023). FaSt-SWOT campaigns. *2023 SWOT Science Team Meeting*, Toulouse, France. Oral. 20 September 2023.

30. Mourre, B., Pascual, A., Balaguer, P., **Barceló-Llull, B.**, *et al.* (2023). FaSt-SWOT multi-platform experiments for SWOT validation in the Western Mediterranean Sea. *2023 SWOT Science Team Meeting*, Toulouse, France. Poster. 19-22 September 2023.

31. **Barceló-Llull, B.**, Pascual, A., Albert, A., Beauchamp, M., Fablet, R., Guinehut, S., Hernández-Lasheras, J., Herrero González, M., Jousset, S., Leroux, S., Mourre, B. & Mulet, S. (2023). T2.3: Observing System Simulation Experiments: impact of multi-platform observations for the validation of satellite observations. *EuroSea Final General Assembly*, Paris, France. Oral (presented by S. Speich). 19 September 2023.
32. **Barceló-Llull, B.** & Pascual, A. (2022). **Invited speaker**. Finescale ocean currents in the Med Sea. *FilaChange workshop*, Paris, France. Oral. 31 August 2022.
33. **Barceló-Llull, B.** & Pascual, A. (2022). Design of in situ sampling strategies to reconstruct fine-scale ocean currents in the context of SWOT satellite validation. *Virtual Poster Session - FilaChange workshop*, Online Everywhere (Twitter). Poster. 26 August 2022.
34. **Barceló-Llull, B.** & Pascual, A. (2022). Design of in situ sampling strategies to reconstruct fine-scale ocean currents in the context of SWOT satellite validation. *EOF-ISMS*, Las Palmas de Gran Canaria, Spain (hybrid). Oral. 6 July 2022.
35. **Barceló-Llull, B.** & Pascual, A. (2022). Evaluating in situ sampling strategies for SWOT satellite validation. *EuroSea/OceanPredict workshop*, Exeter, UK (hybrid). Oral. 29 June 2022.
36. **Barceló-Llull, B.**, Pascual, A., Albert, A., Beauchamp, M., Fablet, R., Hernández-Lasheras, J., Herrero González, M., Leroux, S., Mourre, B., Mulet, S. & Tarry, D. R. (2022). Optimizing multi-platform sampling strategies to anticipate SWOT validation. *ESA Living Planet Symposium 2022*, Bonn, Germany. Poster. 23-27 May 2022.
37. **Barceló-Llull, B.**, Pascual, A., Albert, A., Beauchamp, M., Fablet, R., Guinehut, S., Hernández-Lasheras, J., Herrero González, M., Leroux, S., Mourre, B. & Mulet, S. (2022). T2.3: Observing System Simulation Experiments: impact of multi-platform observations for the validation of satellite observations. *EuroSea Annual Meeting 2022*, Cádiz, Spain. Oral. 9-13 May 2022.
38. **Barceló-Llull, B.**, Pascual, A. & Zarokanellos, N. (2022). Observing system design to validate satellite altimetry: simulated and real observations. *EuroSea Annual Meeting 2022*, Cádiz, Spain. Oral. 9-13 May 2022.
39. **Barceló-Llull, B.** & Pascual, A. (2022). Design of multi-platform sampling strategies for reconstruction of fine-scale ocean currents. *COSS-TT meeting*, Online Everywhere. Oral. 13 April 2022.
40. Pascual, A., **Barceló-Llull, B.** & Cutolo, E. (2021). Summary and advances in inversion activities at IMEDEA. Finescale ocean currents in the Med Sea. *SWOT Science Team Inversion Working Group meeting*, Online Everywhere. Oral. 13 October 2021.

41. **Barceló-Llull, B.**, Pascual, A., Leroux, S., Albert, A., Hernández-Lasheras, J. & Mourre, B. (2021). Optimizing multi-platform sampling strategies to anticipate SWOT validation. *SWOT Science Team regional validation meeting*, Online Everywhere. Oral. 3 June 2021.
42. Tzortzis, R., Doglioli, A. M., Barrillon, S., Petrenko, A. A., d'Ovidio, F., Izard, L., Thyssen, M., Bhairy, N., Pascual, A., **Barceló-Llull, B.**, Cyr, F., Tedetti, M., Garreau, P., Dumas, F., Bordois, L., Comby, C., Rousselet, L. & Gregori, G. (2021). Recent progress in the study of fine-scale physical-biological coupling in the Mediterranean Sea. *EGU General Assembly 2021*, Online Everywhere. vPICO.
43. **Barceló-Llull, B.**, Pascual, A., Speich, S., Cutolo, E., Fablet, R., Gasparin, F., Guinehut, S., Hernández-Lasheras, J., Leroux, S., Mignot, A., Mourre, B., Mulet, S., Rémy, E., & Verbrugge, N. (2020). Optimizing multi-platform sampling strategies to anticipate SWOT validation. *AGU Fall Meeting 2020*, Online Everywhere. Oral.
44. Pascual, A., Fablet, R., Mason, E., Ruiz, S., Sánchez-Román, A., **Barceló-Llull, B.**, Aulicino, G., Cotroneo, Y., Gómez-Navarro, L., Cutolo, E., Rodríguez-Tarry, D., Ouala, S., Antich, H., López-Radcenco, M., Mourre, B., Faugère, Y., Mahadevan, A., Doglioli, A. & d'Ovidio, F. (2020). Multi-platform experiments, numerical simulations and data science techniques for generation of new altimetric products: focus on mesoscale and sub-mesoscale variability (MANATEE - OSTST proposal). *OSTST 2020 Virtual Meeting*, Online. Oral.
45. **Barceló-Llull, B.**, Drushka, K., Gaube, P. & Della Penna, A. (2020). Lagrangian reconstruction of sea surface salinity to extract small-scale variability from SMAP. *Surface currents in the coupled ocean-atmosphere system workshop*, La Jolla, USA. Poster.
46. **Barceló-Llull, B.**, Drushka, K., Gaube, P. & Della Penna, A. (2020). Lagrangian reconstruction of sea surface salinity to extract small-scale variability from SMAP. *Ocean Sciences meeting*, San Diego, USA. Oral.
47. Pascual, A., Ruiz, S., **Barceló-Llull, B.**, Sánchez-Román, A., Mason, E., Cutolo, E., Rodríguez-Tarry, D., Faugère, Y., Gómez-Navarro, L., Ouala, S., Fablet, R., Mourre, B., Tintoré, J., Mahadevan, A. & D'Ovidio, F., (2020). Finescale horizontal and vertical currents from in-situ observations in preparation for SWOT altimeter mission. *Ocean Sciences meeting*, San Diego, USA. Oral.
48. **Barceló-Llull, B.**, Drushka, K., Gaube, P., Rodríguez, E., Morey, S., Chase, A. & Della Penna, A. (2019). Physical and Biological Observations of Submesoscale Ocean Currents. *OceanObs'19*, Hawaii, USA. Poster.
49. Drushka, K., **Barceló-Llull, B.** & Gaube, P. (2019). In situ observations of (sub)mesoscale bio-physical variability for SWOT. *2019 SWOT Science Team Meeting*, Bordeaux, France. Oral.

50. Pascual, A., **Barceló-Llull, B.**, Cutolo, E., Sánchez-Román, A., Díaz-Barroso, L., Alou, E., Mourre, B., Allen, J., Muñoz, C., Hernández, J.-G., Antich, H., Gómez-Navarro, L., Torner, M., Ruiz, S., Mason, E., Ser-Giacomi, E., Cyr, F., Doglioli, A., Dumas, F., & D'Ovidio, F., (2019). The PRE-SWOT experiment. *2019 SWOT Science Team Meeting*, Bordeaux, France. Oral.

51. **Barceló-Llull, B.**, Pascual, A., Ruiz, S., Escudier, R., Torner, M. & Tintoré, J. (2019). Temporal and Spatial Hydrodynamic Variability in the Mallorca Channel (Western Mediterranean Sea) From 8 Years of Underwater Glider Data. *8th EGO Meeting and International Glider Workshop*, New Brunswick, United States. Oral.

52. Pascual, A., Ruiz, S., Sánchez-Román, A., Gómez-Navarro, L., **Barceló-Llull, B.**, Díaz-Barroso, L., Chabert, P., Cutolo, E., Freilich, M. A., Heslop, E., Casas, B., Torner, M., Mourre, B., Alou, E., Cotroneo, Y., Aulicino, G., Mason, E., Mahadevan, A., Tintoré, J., D'Ovidio, F., Fablet, R. & Allen, J. (2018). Review: Overview of Fine-scale Multiplatform Experiments in the Southwest Mediterranean Sea: Lessons Learnt in the Last Five Years. *25 Years of Progress in Radar Altimetry Symposium*, Azores, Portugal. Oral.

53. Pascual, A., **Barceló-Llull, B.**, Gómez-Navarro, L., Díaz-Barroso, L., Casas, B., Muñoz, C., Mourre, B., Torner, M., Ruiz, S., Allen, J., D'Ovidio, F., Doglioli, A. & Dumas, F. (2018). PRE-SWOT multi-platform experiment in the SouthWest Mediterranean Sea. *Coastal Ocean and Shelf Seas Task Team (COSS-TT) - GODAE OceanView*, Madrid, Spain. Poster.

54. **Barceló-Llull, B.**, Díaz-Barroso, L., Casas, B., Gómez-Navarro, L., Cutolo, E., Sánchez-Román, A., Escudier, R., Cabornero, A., Calafat, N., Capó, E., Ser-Giacomi, E., Mason, E., Mourre, B., Muñoz, C., Ruiz, I., Alou, E., Ruiz, S., Torner, M., Miralles, A., Rubio, M., Vizoso, G., Tintoré, J., Allen, J. & Pascual, A. (2018). Characterizing the Southwest Mediterranean Sea to anticipate the SWOT satellite launch. *Encuentro Oceanografía Física Española*, Vigo, Spain. Poster.

55. **Barceló-Llull, B.**, Pascual, A., Ruiz, S., Allen, J. & Torner, M. (2018). Temporal and spatial variability of the Mallorca channel from underwater glider data. *EGU General Assembly*, Vienna, Austria. PICO.

56. **Barceló-Llull, B.**, Mason, E., Capet, A. & Pascual, A. (2016). Impact of vertical and horizontal advection on nutrient distribution in the southeast Pacific. *New era of altimetry, new challenges*, La Rochelle, France. Poster.

57. **Barceló-Llull, B.**, Sangrà, P., Pallàs-Sanz, E., Barton, E. D., Estrada-Allis, S. N., Martínez-Marrero, A., Aguiar-González, B., Grisolía, D., Gordo, C., Rodríguez-Santana, A., Marrero-Díaz, A. & Arístegui, J. (2016). Anatomy of a subtropical mode water Eddy. *Encuentro Oceanografía Física Española*, Alacant, Spain. Oral. *Award for best oral communication of students (Organizing Committee).

58. Martínez-Marrero, A., Sangrà, P., Estrada-Allis, S. N., Rodríguez-Santana, A., Aguiar-González, B., Marrero-Díaz, A., Barton, E. D., Pallàs-Sanz, E., **Barceló-Llull, B.**, Gordo, C., Grisolía, D. & Cana, L. (2016). Near-inertial waves trapping and deep mixing in an open ocean anticyclonic eddy. *Encuentro Oceanografía Física Española*, Alacant, Spain. Oral.

59. **Barceló-Llull, B.**, Pallàs-Sanz, E. & Sangrà, P. (2016). Inference and Biogeochemical Response of Vertical Velocities inside a Mode Water Eddy. *Ocean Sciences Meeting*, New Orleans, United States. Oral.

60. Pascual, A., Gaube, P., Mason, E., **Barceló-Llull, B.** & Ruiz, S. (2016). Ocean mesoscale eddies and vertical motion. *Ocean Sciences Meeting*, New Orleans, United States. Poster.

61. **Barceló-Llull, B.**, Pallàs-Sanz, E. & Sangrà, P. (2015). Inference of vertical velocities in a mid ocean anticyclonic mesoscale eddy. *International Union of Geodesy and Geophysics (IUGG)*, Praga, Czech Republic. Oral.

62. Sangrà, P., **Barceló-Llull, B.**, Estrada-Allis, S. N., Martínez-Marrero, A., Aguiar-González, B., Grisolía, D., Gordo, C., Ramírez-Garrido, S., Pallàs-Sanz, E., Rodríguez-Santana, A., Marrero-Díaz, A., Arístegui, J., Barton, E. D. & Cana, L. (2015). Enhancement of primary production and deep mixing by a mesoscale mid ocean anticyclonic Eddy. *International Union of Geodesy and Geosphysics (IUGG)*, Praga, Czech Republic. Oral.

63. Martínez-Marrero, A., Sangrà, P., **Barceló-Llull, B.** & Gordo, C. (2015). Near-inertial waves trapping by a mid ocean anticyclonic eddy. *International Union of Geodesy and Geosphysics (IUGG)*, Praga, Czech Republic. Poster.

64. Marrero-Díaz, A., Gordo, C., Sangrà, P., Rodríguez-Santana, A., Estrada-Allis, S. N., Aguiar-González, B. & **Barceló-Llull, B.** (2015). Mixing of water masses in an intrathermocline Eddy. *International Union of Geodesy and Geosphysics (IUGG)*, Praga, Czech Republic. Poster.

65. Grisolía-Santos, D., Rodríguez-Cruz, E., Sangrà, P., Martínez-Marrero, A., Cana, L. & **Barceló-Llull, B.** (2015). Lagrangian evolution of a mid ocean anticyclonic eddy. *International Union of Geodesy and Geosphysics (IUGG)*, Praga, Czech Republic. Poster.

66. **Barceló-Llull, B.**, Mason, E. & Pascual, A. (2014). Quasi-geostrophic vertical motion from satellite and in-situ observations: Impact on South East Pacific nitrate distribution through a Lagrangian simulation. *Encuentro Oceanografía Física Española*, Las Palmas de Gran Canaria, Spain. Oral. *Registration grant (Organizing Committee).

67. **Barceló-Llull, B.**, Mason, E. & Pascual, A. (2013). Applications of Mesoscale Dynamics: Impacts of vertical motion on nitrate distribution. *EGU General Assembly*, Vienna, Austria. Poster.

68. Rio, M.-H., Pascual, A., Poulain, P.-M., Menna, M., Adani, M., Balbin, R., Aparicio, A., López Jurado, J. L., **Barceló, B.** & Tintoré, J. (2013). A new Mean Dynamic Topography of the Mediterranean Sea based on model outputs, drifter data, hydrological profiles and altimeter measurements. *EGU General Assembly*, Vienna, Austria. Poster.
69. **Barceló-Llull, B.**, Mason, E. & Pascual, A. (2012). Applications of mesoscale dynamics: Impacts of vertical motion on nitrate distribution. *Encuentro Oceanografía Física Española*, Madrid, Spain. Poster.

Student supervision

1. **Blanca Fernández Álvarez** (in progress). PhD thesis: *Surface and Subsurface Marine Heatwaves in the Mediterranean Sea.* Universitat de les Illes Balears. Granted an FPU fellowship. Supervisors: A. Pascual, **B. Barceló-Llull**.
2. **Adriana Socorro Alonso** (in progress). Master's thesis: *Decadal trends in mesoscale eddy activity in the Canary Eddy Corridor*. Master's Degree in Oceanography. Universidad de Las Palmas de Gran Canaria. Supervisors: Yeray Santana-Falcón, Nauzet Hernández-Hernández, **B. Barceló-Llull**.
3. **Paula Alós Lluesma** (15/09/2025; 5.0). Master's thesis: *Long-Term data series from Surface Ocean Eddy Kinetic Energy Using Satellite Altimetry: a 30-Year Case Study in the North Atlantic Ocean.* Master Erasmus Mundus – Marine Environment and Resources (MER2030). UPV/EHU. Supervisors: A. Pascual, **B. Barceló-Llull**.
4. **Cristina Martí Solana** (03/10/2024; 9.7). Master's thesis: *Detection of salinity fronts from satellite observations.* Master's Degree in Advanced Physics and Applied Mathematics. Universitat de les Illes Balears. Supervisors: V. Combes, **B. Barceló-Llull**.
5. **Blanca Fernández Álvarez** (28/06/2024; 10 MH). Master's thesis: *Marine Heat Waves in the Balearic Sea.* Master's Degree in Oceanography. Universidad de Las Palmas de Gran Canaria. Supervisors: A. Pascual, **B. Barceló-Llull**.
6. **Marta Veny López** (September-October 2023). Supervision of a PhD student stay at IMEDEA. Home Institution: Instituto de Oceanografía y Cambio Global (IOCAG) – Departamento de Física, Universidad de las Palmas de Gran Canaria (ULPGC). Supervisors of the stay: S. Ruiz, **B. Barceló-Llull**.
7. **Elisabet Verger Miralles** (14/09/2023; 9.6). Master's thesis: *Mesoscale ocean structure reconstruction through data collection and analysis in the framework of the FAST-SWOT project.* Master's Degree in Advanced Physics and Applied Mathematics. Universitat de les Illes Balears. Supervisors: A. Pascual, **B. Barceló-Llull**, A. Amores.

8. **Helena Antich Homar** (19/10/2022; 9.5). Master's thesis: *Procesado de datos y entrenamiento de la red neuronal de un sistema IoT para predecir las trayectorias de Pygoscelis Antarcticus a partir de variables oceanográficas.* Master's degree in Intelligent Systems. Universitat de les Illes Balears. Supervisors: **B. Barceló-Llull**, A. Pascual, M. Barranco.

Teaching experience

1. **Teaching assistant** at the Universitat de les Illes Balears, Spain. 20353 - Mechanics (20 hours). Degree in Construction. Academic year 2024-25. `https://www.uib.eu/Learn/estudis-de-grau/grau/edificacio/GEED-P/20353/`.
2. **Teaching assistant** at the Universitat de les Illes Balears, Spain. 20356 - Fundamentals of Facilities (43 hours). Degree in Construction. Academic year 2023-24. `https://www.uib.eu/Learn/estudis-de-grau/grau/edificacio/GEED-P/20356/`.
3. **Guest lecturer** at the Universitat de les Illes Balears (UIB), Spain. 21054 - Physical Oceanography. Degree in Physics. Academic year 2022-23. Delivered one lecture of 2 hours by invitation.
4. **Certification PROFESORA CONTRATADA DOCTORA**. Agencia Nacional de Evaluación de la Calidad y Acreditación (ANECA). September 2025.
5. **Certification PROFESORA DE UNIVERSIDAD PRIVADA**. Agencia Nacional de Evaluación de la Calidad y Acreditación (ANECA). September 2025.
6. **Certification PROFESORA AYUDANTE DOCTORA**. Agencia Nacional de Evaluación de la Calidad y Acreditación (ANECA). November 2022.

Outreach activities

1. *Type of activity:* Live video on YouTube. International Women's Day.
   Title of the activity: ¿Abandonar o resistir? Maternidad y conciliación en la carrera científica
   Role: Moderator.
   Place: YouTube.
   Date: 05/03/2025
   Link: `https://www.youtube.com/live/2xIq23p3T9Q`
2. *Type of activity:* Interview in a newspaper.
   Title of the activity: Bàrbara Barceló-Llull: "Quan vaig entrar a la Universitat de Washington no podia deixar de pensar amb els padrins".
   Role: Interviewed.
   Place: Tot Pla.
   Date: 01/12/2024
   Link: `https://www.totpla.cat/2024/12/01/barbara-barcelo-llull-quan-vaig-entrar`

3. *Type of activity:* FaSt-SWOT documentary.
   Title of the activity: Del espacio al Mediterráneo: Persiguiendo corrientes marinas.
   Role: Participation, revision.
   Place: YouTube.
   Date: 09/11/2023
   Link: https://youtu.be/WQd9LeIdLSk?si=EfF_xSEMSMhKQ-ex
4. *Type of activity:* Presentation of the FaSt-SWOT documentary.
   Title of the activity: Estreno del corto documental "Del espacio al Mediterráneo: Persiguiendo corrientes marinas".
   Place: IMEDEA.
   Date: 08/11/2023
   Link: https://imedea.uib-csic.es/divulgacion-y-comunicacion/noticias/?new_id=2041
5. *Type of activity:* Live video on YouTube.
   Title of the activity: Directo proyecto FaSt-SWOT - Día Mundial de los Océanos.
   Role: Original idea, scriptwriting, organization, moderator, dissemination on social media.
   Place: YouTube.
   Date: 08/06/2023
   Link: https://www.youtube.com/live/__mim-lSua4?feature=share
6. *Type of activity:* Stories on Instagram.
   Title of the activity: Dissemination of the FaSt-SWOT experiment.
   Place: Instagram.
   Date: 25-27/04/2023
   Link: https://www.instagram.com/stories/highlights/17965615391215848/
7. *Type of activity:* Tweets on Twitter.
   Title of the activity: Dissemination of the FaSt-SWOT experiment.
   Place: Twitter.
   Date: 25/04/2023 and 27/04/2023
   Links: https://twitter.com/bbarcelo_ocean/status/1650837240354947072, https://twitter.com/bbarcelo_ocean/status/1650972225502076932, https://twitter.com/bbarcelo_ocean/status/1651548870868971521
8. *Type of activity:* Interview.
   Title of the activity: The new wave of oceanographers: Bàrbara Barceló-Llull.
   Place: SWOT-AdAC website.
   Date: 21/04/2023
   Link: https://www.swot-adac.org/news/the-new-wave-of-oceanographers-barbara-ba

9. *Type of activity:* Taller para la Ocean Night.
   Title of the activity: Moviéndonos en el océano.
   Place: IMEDEA.
   Date: 01/12/2022
10. *Type of activity:* Talleres para la Semana de la Ciencia y la Tecnología.
   Title of the activity: Moviéndonos en el océano.
   Place: IMEDEA.
   Date: 25/11/2022 and 29/11/2022
   Link: https://imedea.uib-csic.es/divulgacion-y-comunicacion/noticias/?new_id=1991
11. *Type of activity:* Tweet on Twitter.
   Title of the activity: Report: Recommendations for the design of in situ experiments to validate SWOT!!
   Place: Twitter.
   Date: 15/11/2022
   Link: https://twitter.com/bbarcelo_ocean/status/1592509065963663361
12. *Type of activity:* Virtual Poster on Twitter.
   Title of the activity: Virtual Poster Session during the FilaChange workshop
   Place: Twitter.
   Date: 26/08/2022
   Link: https://twitter.com/SWOT_AdAC/status/1563197631635292161
13. *Type of activity:* Tweet on Twitter.
   Title of the activity: Open and free codes to evaluate in situ sampling strategies aimed at reconstructing fine-scale ocean currents
   Place: Twitter.
   Date: 05/07/2022
   Link: https://twitter.com/bbarcelo_ocean/status/1544244579439988737
14. *Type of activity:* Documentary, International Day of Women and Girls in Science at IMEDEA.
   Title of the activity: Ilusiones científicas.
   Place: Published on YouTube.
   Date: 11/02/2021
   Link: https://youtu.be/IVexfmEDMRg

15. *Type of activity:* Microcharlas virtuales con científicos/as, CSIC Science and Technology Week.
Title of the activity: Els corrents marins: Què són? Per què són importants? Com els podem observar?
Place: online.
Date: 11/11/2020
Link: `https://youtu.be/bGwi7niAqno`

Merits, achievements and experience

RESEARCH NETWORKS

Member of IAPSO Best Practice Study Group on Numerical Methods in Computational Lagrangian Oceanography, 2026-2028.

Member of the ISSI International Team "Relationship between the Atlantic meridional overturning circulation and coastal sea level: New insights from next-generation satellite observations", 2025-2027.

Member of the SWOT Science Team, *Co-PI VERSO project*, 2020-present.

Member of LINCC UIB (Laboratori Interdisciplinari sobre Canvi Climàtic), 2025-present.

RESEARCH STAYS

Departamento de Matemática Aplicada e Ingeniería Aeroespacial de la Universidad de Alicante, *Spain*, 1 February 2025 - 31 July 2025, 6 months.
Supervisors: M. Isabel Vigo Aguiar, Juan Manuel Sayol

Centro de investigación científica y de educación superior de Ensenada, CICESE, *México*, 2016, 120 days.
Pre-doctoral mobility grant from the Spanish National Research Program (EEBB-I-15-09833). Supervisor: Enric Pallàs-Sanz

Centro de investigación científica y de educación superior de Ensenada, CICESE, *México*, 2015, 120 days.
Pre-doctoral mobility grant from the Spanish National Research Program (EEBB-I-16-11015). Supervisor: Enric Pallàs-Sanz

Institut Mediterrani d'Estudis Avançats, IMEDEA, *Spain*, 2015, 30 days.
Supervisor: Ananda Pascual

CRUISES

FaSt-SWOT, *Fine-Scale ocean currents from integrated multi-platform experiments and numerical simulations: contribution to the new SWOT satellite mission*, R/V García del Cid, April-May 2023, 8 days.

BioSWOT-Med, *Biological applications of the satellite Surface Water and Ocean Topography in the Mediterranean*, R/V L'Atalante, April-May 2023, 25 days.

PRE-SWOT, *Mesoscale and sub-mesoscale vertical exchanges from multi-platform experiments and supporting modeling simulations: anticipating SWOT launch*, R/V García del Cid, May 2018, 12 days.

PUMP, *Study of the Vertical Oceanic Pump in mesoscale eddies*, R/V Hespérides, September 2014, 22 days.

COMMISSION PARTICIP.

Member of the Scientific Committee of the 57th International Liège Colloquium, 2026.

Member of the Scientific Committee of the VIII Encuentro Oceanografía Física conference (EOF), 2024.

Member of the Commission for communication and scientific dissemination at IMEDEA (CSIC-UIB), 2020-2021.

SESSION ORGANISING

Unraveling Physical-Biological Interactions at Meso- and Submesoscales, *Ocean Sciences Meeting, Glasgow, Scotland*, 2026.

The realm of (sub)mesoscale dynamics: variability, impact, and new challenges, *EGU General Assembly, Vienna, Austria*, 2019.

SESSION CHAIRING

Session 2.4 - Open Ocean, *30 Years of Progress in Radar Altimetry Symposium, Montpellier, France*, 2024.

Session 2.3 - Open Ocean, *30 Years of Progress in Radar Altimetry Symposium, Montpellier, France*, 2024.

Oral Session 5a. Ocean methods: instrumentation, remote sensing and numerical methods, *VIII Encuentro Oceanografía Física conference (EOF 2024), Valencia, Spain*, 2024.

Oral Session 6a. Operational oceanography and marine weather, *VIII Encuentro Oceanografía Física conference (EOF 2024), Valencia, Spain*, 2024.

PEER REV.

Nature Geoscience, 2025, 2025.

Nature Communications, 2023.

Ocean Science, 2023.

Journal of Geophysical Research: Oceans, 2024, 2022, 2019.

Geophysical Research Letters, 2020.

Journal of Marine Systems, 2019.

Ocean Dynamics, 2019, 2018.

GUEST EDITOR

Special Issue: "Applications of Satellite Altimetry in Ocean Observation", *Remote Sensing*, 2023-2025, `https://www.mdpi.com/journal/remotesensing/special_issues/OT3DF5K943`.

Special Issue: "Application of Remote Sensing for the Study of Coastal and Shelf Seas Dynamics", *Remote Sensing*, 2023-2024, `https://www.mdpi.com/journal/remotesensing/special_issues/6T38DXQ3Y5`.

ADDITIONAL TRAINING

GODAE OceanView International School, *Mallorca, Spain*, 2-13 October 2017.

*Registration and accommodation grant (GOV International School directors).

International workshop FineMed 2017, *Marseilles, France*, 26-28 June 2017, Scientific Committee: F. d'Ovidio (LOCEAN, Paris, France), A. Doglioli (MIO, Marseille, France), G. Grégori (MIO, Marseille, France), P. Testor (LOCEAN, Paris, France), B. Fach (METU, Erdemli, Turkey).
*Travel grant (SWOT Science Team).

A Short Course on Time Series Analysis, *UIB, Mallorca, Spain*, 26-30 Jan 2015, Prof. Dudley B. Chelton (CEOAS, Oregon State University, USA).

Initiation to physical oceanography , *Institut Mediterrani d'Estudis Avançats, IMEDEA, Mallorca, Spain*, 24 May 2012 -12 Oct 2012, 440 hours, Supervisors: Ananda Pascual and Evan Mason.

Técnicas avanzadas para el tratamiento numérico y representación gráfica de datos (7a edición), *Universidad de Salamanca, Spain*, 2011, 40 hours.

Presentaciones y comunicaciones orales: estrategias y nuevas tecnologías (6a edición), *Universidad de Salamanca, Spain*, 2011, 40 hours.

Skills

LANGUAGES: English – fluent

Spanish – native speaker

Catalan – native speaker